WONBUILDING

Agenda

Project Overview

Problem & Solutions

Technology

Final Deliverable

Impact

OVERVIEW

Project Overview

Project Information

Name	Commercial Real Estate Data Pipeline and Dashboard Development
Duration	2023.04 ~ 2024.12
Budget	20M KRW grant
Team	3 members
Summary	Based on consultant workflow analysis, built an automated ETL pipeline that unmasks anonymized transactions from fragmented commercial data with monthly automated delivery and developed a real-time dashboard for analytics. This transforms traditional manual, intuition-based consulting into automated, data-driven operations, establishing Wonbuilding as the industry leader.
Objectives	Market Leadership Positioning Operational Workforce Optimization Process Automation & Efficiency Data Integration & Transparency Real-time Market Intelligence

Client Information

WONBUILDING

Comprehensive Commercial Real Estate Consulting Company

Building Transactions
Sales & leasing consulting for commercial properties

Development Consulting
Feasibility studies & business site analysis

Asset & Facility Management
Property, financial, and legal administration

Corporate Support
Office operations, procurement, and external affairs management

Key Metrics

#1
Real Estate Consulting Company

24
Years of experience

200,000+
Accumulated clients served

100,000+
Commercial properties managed

PROBLEMS & SOLUTIONS

Problems & Solutions

As-Is

Limited Data Accessibility

- Fragmented sources across platforms (MOLIT, Building Hub, V-World)
- High collection & cleaning costs
- Core commercial real estate datasets restricted in scope and format

Difficulty Tracking Market Trends

- No integrated visualization of transactions & 5 years price trends
- Reliance on experience or intuition over data

Lack of Market Transparency

- Absence of standardized analytics makes it harder to compare districts and assess investment opportunities
- Clients cannot easily access reliable, consolidated information

To-be

- Integrated Data Modeling & Process Standardization
- Real-time Market Intelligence Dashboard
- Automated ETL Pipelines

Expected Impact

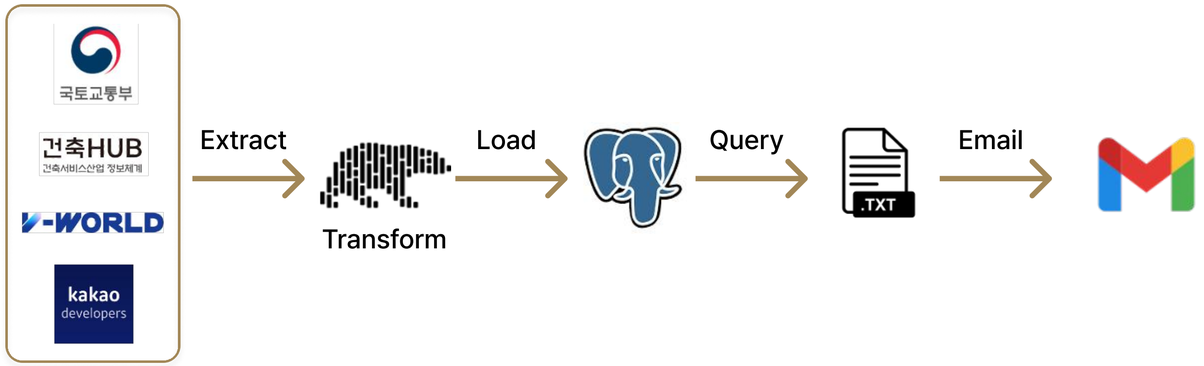
- Conversion into **long-term digital assets**
- Operational **workforce optimization**
- Data-driven **decision-making enhancement**

TECHNOLOGY

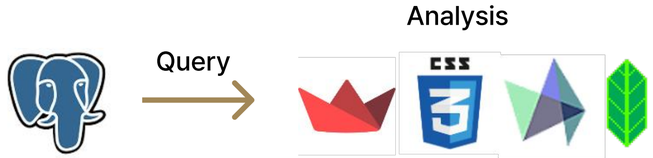
Technology Architecture

Infrastructure

Data Pipeline



Website



TECHNOLOGY

Data Pipeline

Utilize Apache Airflow to orchestrate ETL workflows

Extract data from four distinct sources (PythonOperators)

- Real Estate Transaction Records
- Building Registries
- Land Cadastral Data
- Geospatial Location Data

Aggregated and transformed using Polars to identify buildings associated with masked transaction records (PythonOperators)

Employ PostgreSQL structured with a dual-tiered schema (PostgresOperator)

- Original data is stored in a dedicated raw_data schema
- Refined data is stored in a preprocessed_data schema

Package the most recent three months' data into a text file and delivered to our client's email (EmailOperator)

Website

Provides real estate consultants tailored insights to inform their business decisions

Utilize Nginx as a reverse proxy to restrict access based on a predefined IP whitelist ensuring that the website is exclusively available to internal company users

Provides a hierarchical data analysis framework, allowing users to explore insights at three distinct scopes

- District
- Sub District
- Individual Building

Built front-end using Streamlit that enables interactive data exploration

Integrated Highcharts, Folium to generate dynamic visualizations

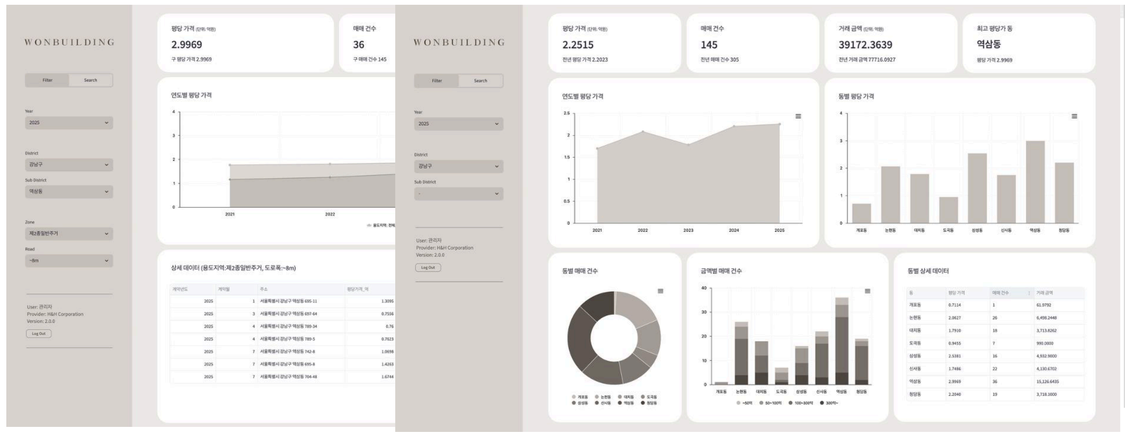
Styled using CSS to ensure professional look

FINAL DELIVERABLE

Final Ouput Overview

Real estate transaction data sent via email

Our data pipeline ensures that clients
receive timelytransaction data.
It automatically packages the most recent
three months of data into a secure text file
and delivers it via email every month, providing a reliable
source of information without any manual effort.



Interactive Dashboard for Market Insights

Our visualization dashboard allows users to explore
district-level property data with ease.
It provides interactive charts, trend analyses, and search
functionality, enabling sales representatives to identify
market opportunities quickly and support clients with
data-driven recommendations.

FINAL DELIVERABLE

Dashboard

Real-time data analysis dashboard (District)

Features

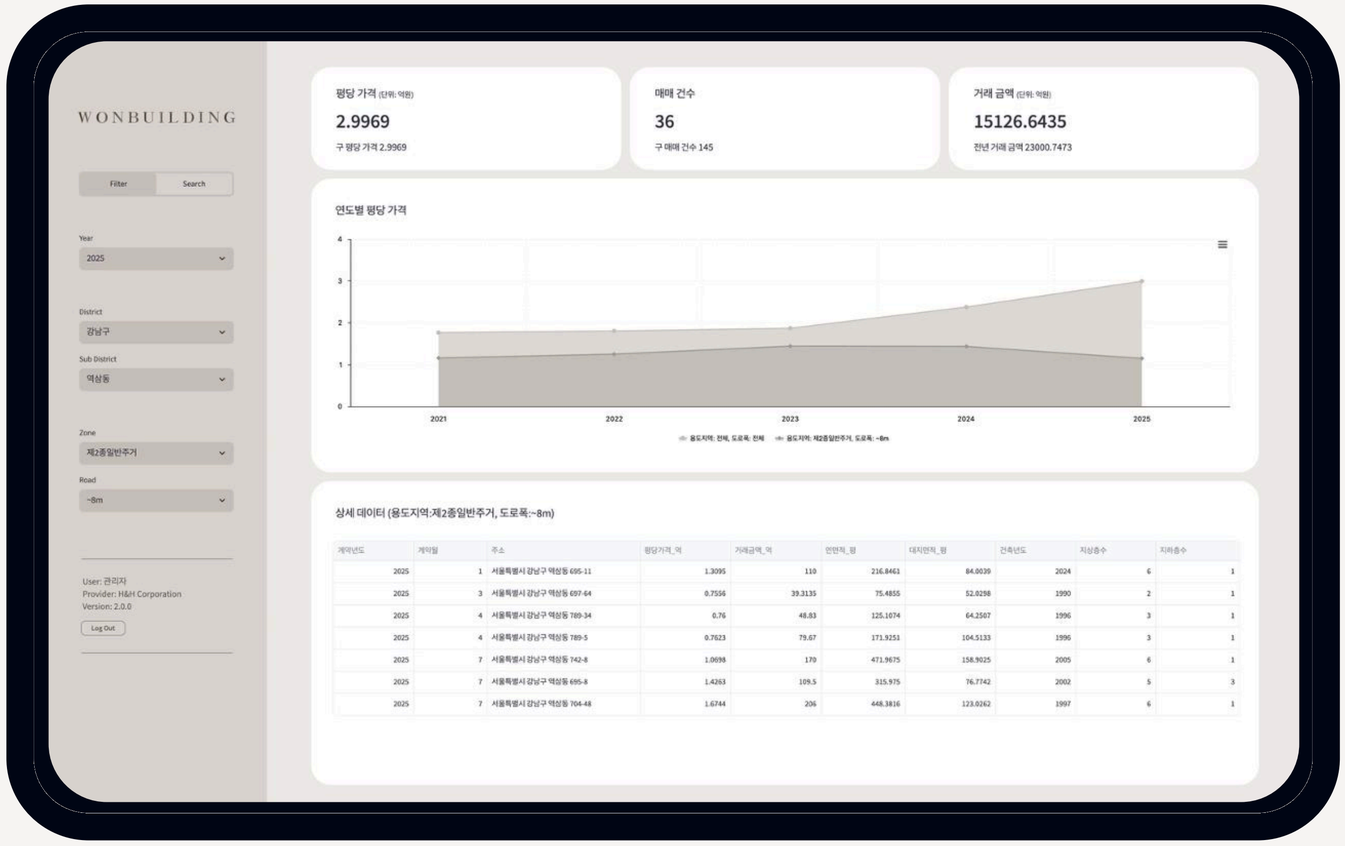
- Aggregated metrics for the entire district (avg. price, transaction volume, total deal value)
- Cross-neighborhood comparisons through bar and donut charts
- **Yearly trend analysis** of district-level prices and volumes
- **Deal size** distribution by price range

Impact

- Provides a **macro-level view** of commercial property dynamics
- Helps identify **emerging hotspots** across neighborhoods
- Supports sales teams in presenting **district-wide market summaries** to clients

FINAL DELIVERABLE

Dashboard



Real-time data analysis dashboard (Sub-District)

Features

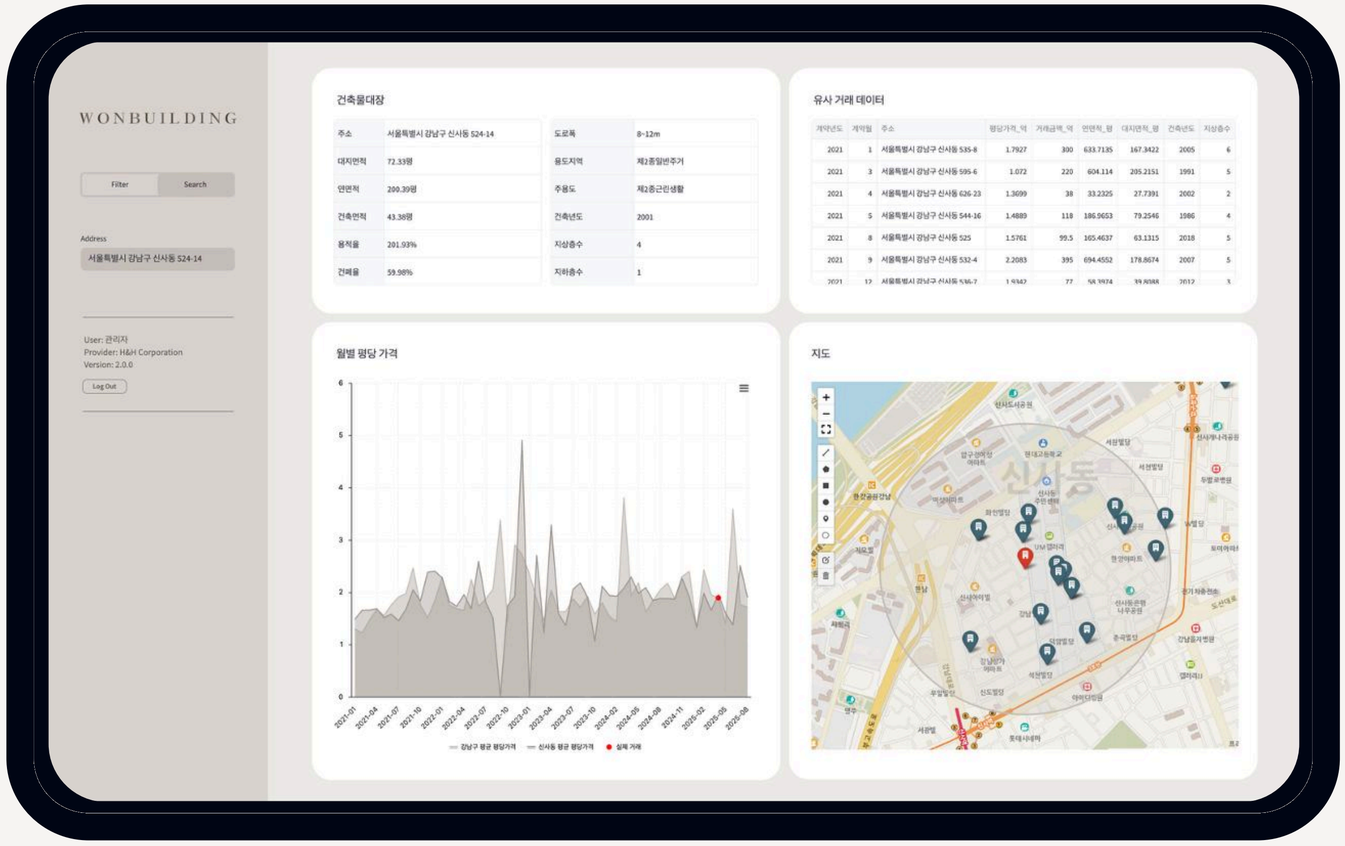
- Focused metrics for a single neighborhood (avg. price, transaction volume, deal value)
- Yearly price trend visualization at the neighborhood level
- **Advanced filtering** (zoning, road width, etc.) for granular analysis
- **Detailed transaction table** with property-level attributes (address, size, year built, floors)

Impact

- Enables **micro-level insights** into specific local markets
- Facilitates **property-level decision-making** with precise data
- Enhances client trust by delivering **transparent, verifiable** information

FINAL DELIVERABLE

Dashboard



Real-time data analysis dashboard (Building-Level)

Features

- Detailed attributes of the selected building (lot size, floor area, building age, zoning, road width, etc.)
- **Recent deals for similar buildings** in the same neighborhood, with unit price, lot size, and building specs.
- Visualization of average **unit price trends** compared against district and neighborhood benchmarks.
- Geospatial visualization of **comparable properties** and recent transactions.

Impact

- Provides a **building-specific** analysis that supports client negotiations.
- Reduces manual effort by integrating property details, comparable data, and trends into one view.
- Helps sales representatives deliver **data-backed recommendations** on individual properties.

IMPACT

Impact

01 Data Processing Efficiency

Manual data collection time reduced by 80% through ETL automation

02 Market Coverage

Consolidated 100,000+ property records across Seoul districts

03 Consultant Confidence Improvement

Achieving 45% consultant confidence improvement

04 Client Engagement

Transparent dashboards expected to improve client trust by 30%